Lecture #15 | Machine learning (cont.)

SE377 Introduction to Big Data Analysis and Visualization (2017)

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Testing and Evaluation

- Cross-validation
- Evaluation metrics
  - confusion matrix
  - F-score
  - Decision threshold
- Dummy classifier
- Multi-class classification and evaluation
- Model selection

Recap: evaluation metrics with confusion matrix

True negative	**TN**	**FP**
True positive	**FN**	**TP**
	Predicted negative	Predicted positive

- $\text{accuracy} = \frac{\text{TN}+\text{TP}}{\text{TN}+\text{TP}+\text{FN}+\text{FP}}$
- $\text{classfication error} = \frac{\text{FP}+\text{FN}}{\text{TN}+\text{TP}+\text{FN}+\text{FP}}$
- $\text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}$
  - true positive rate (TPR)
  - sensitivity
  - probability of detection
- $\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$
- $\text{false positive rate} = \frac{\text{FP}}{\text{TN}+\text{FP}}$
  - specificity

Decision Functions

- Each classifier score value per test point indicates how confidently the classifier predicts the positive class (large-magnitude positive values) or the negative class (large-magnitude negative values)
- Choosing a fixed decision threshold gives a classification rule
- By sweeping the decision threshold through the entire range of possible score values, we get a series of classification outcomes that form a curve

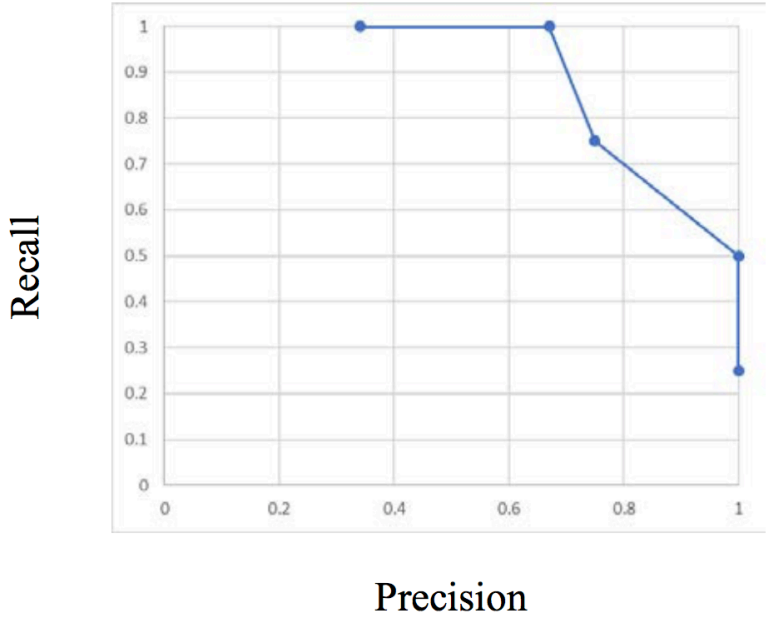
Predicted Probability of Class Membership

- Typical rule: choose most likely class
  - e.g., class 1 if threshold > 0.50
- Adjusting threshold affects predictions of classifier
- Higher threshold results in a more conservative classifier
  - e.g., only predict Class 1 if estimated probability of class 1 is above 70%
  - This increases precision. Doesn't predict class 1 as often, but when it does, it gets high proportion of class 1 instances correct
- Not all models provide realistic probability estimates

Varying the Decision Threshold

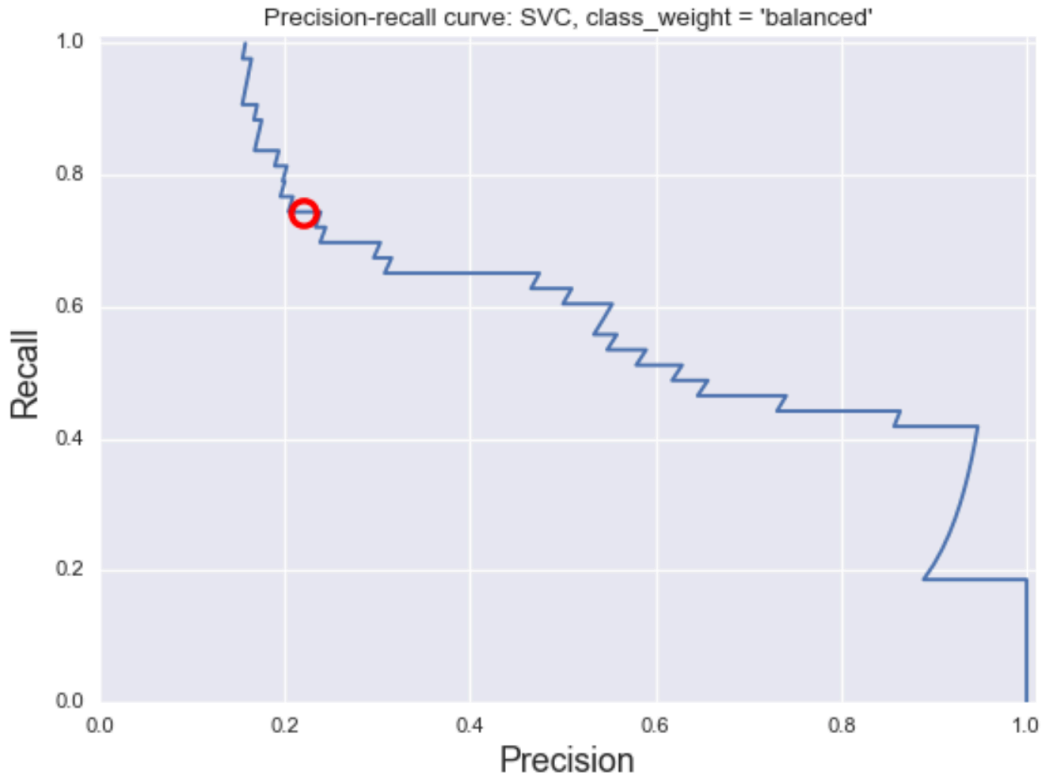
True Label	Classifier score
0	-27.6457
0	-25.8486
0	-25.1011
0	-24.1511
0	-23.1765
0	-22.575
0	-21.8271
0	-21.7226
0	-19.7361
0	-19.5768
0	-19.3071
0	-18.9077
0	-13.5411
0	-12.8594
1	-3.9128
0	-1.9798
1	1.824
0	4.74931
1	15.234624
1	21.20597

Classifier score threshold	Precision	Recall
-20	4/12=0.34	4/4=1.00
-10	4/6=0.67	4/4=1.00
0	3/4=0.75	3/4=0.75
10	2/2=1.0	2/4=0.50
20	1/1=1.0	1/4 = 0.25



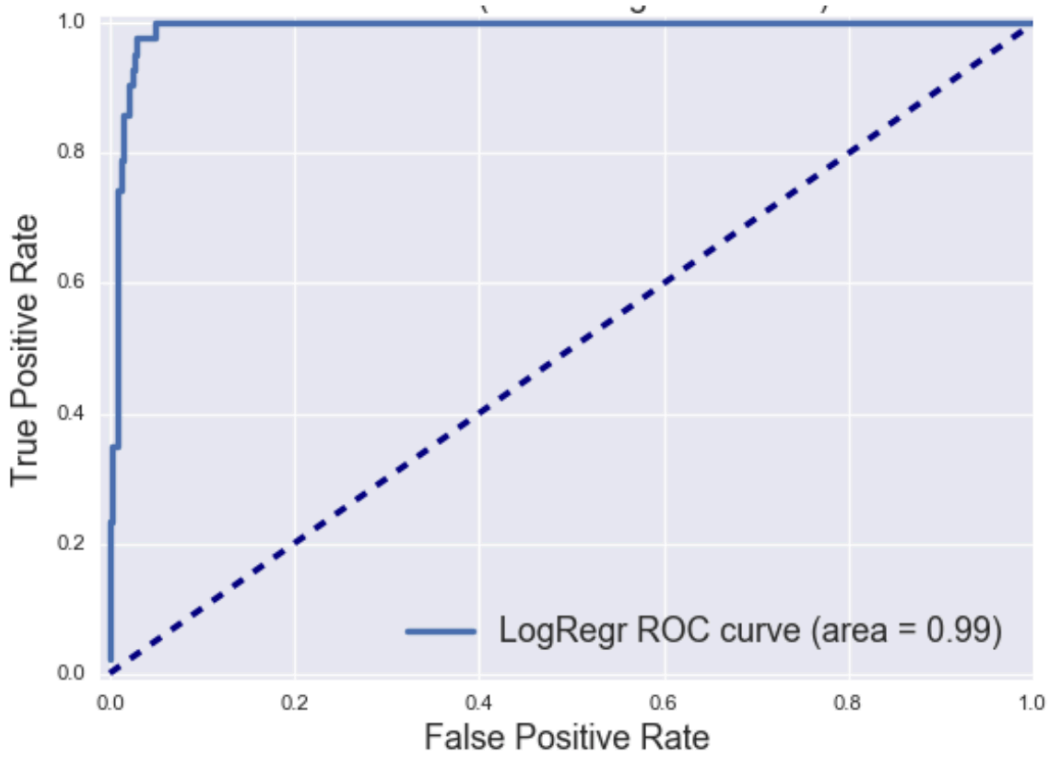
Precision-Recall curve

- X-axis: Precision Y-axis: Recall
- Top right corner:
  - The “ideal” point
  - Precision = 1.0
  - Recall = 1.0
- “Steepness” of P-R curves is important
  - Maximize precision
  - while maximizing recall



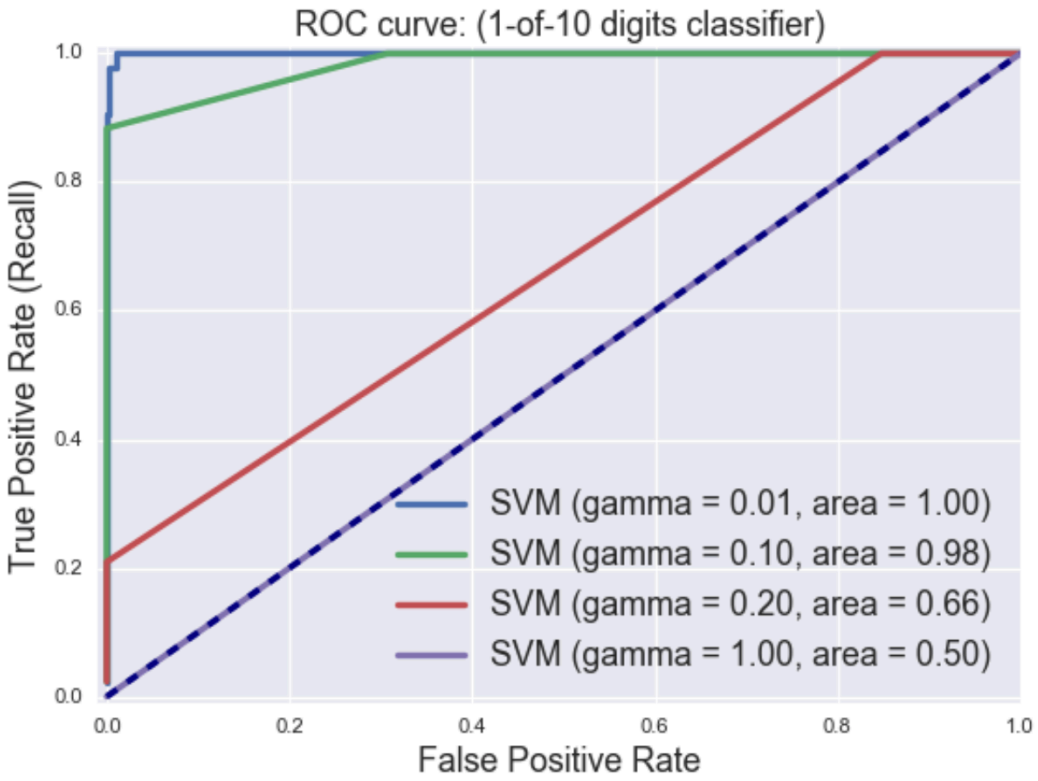
ROC curve

- X-axis: False Positive Rate Y-axis: True Positive Rate
- Top left corner
  - The “ideal” point
  - False positive rate of zero
  - True positive rate of one
- “Steepness” of ROC curves is important
  - Maximize the true positive rate
  - While minimizing the false positive rate



Area Under the Curve (AUC)

- AUC = 0 (worst) AUC = 1 (best)
- AUC can be interpreted as
  - The total area under the ROC curve
  - The probability that the classifier will assign a higher score to a randomly chosen positive example than to a randomly chosen negative example
- Advantages
  - Gives a single number for easy comparison
  - Does not require specifying a decision threshold.
- Drawbacks
  - As with other single-number metrics, AUC loses information, e.g., about tradeoffs and the shape of the ROC curve
  - This may be a factor to consider when e.g. wanting to compare the performance of classifiers with overlapping ROC curves



Dummy classifier

- What accuracy/recall/precision is good enough?
  - e.g., 0.9? 0.99? 0.999?
- Dummy classifiers serve as a sanity check on your classifier's performance
- They provide a null metric (e.g. null accuracy) baseline
- Dummy classifiers should not be used for real problems

Dummy classifiers completely ignore the input data!

- Some commonly-used settings for the `strategy` parameter for DummyClassifier in scikit-learn
  - most_frequent: predicts the most frequent label in the training set
  - stratified: random predictions based on training set class distribution
  - uniform: generates predictions uniformly at random
  - constant: always predicts a constant label provided by the user
- A major motivation of this method is F1-scoring, when the positive class is in the minority.

Multi-class classification

- A SVM is a binary classifier, but many real-world problems have more than two classes
- To get M-class classifier, construct a set of binary classifiers $f^1, f^2, \ldots, f^M$, each trained to separate one class from the rest
- Combine them to get a multi-class classification according to the maximal output before applying the sgn function

$$\underset{j=1\ldots M}{\operatorname{argmax}}\, g^j(x)\,, \text{where } g^j(x) = \sum_{i=0}^{m} y_i \alpha_i^j k(x, x_i) + b^j$$

Multi-class classification (cont.)

$$\underset{j=1\ldots M}{\operatorname{argmax}}\, g^j(x)\,, \text{where } g^j(x) = \sum_{i=0}^{m} y_i \alpha_i^j k(x, x_i) + b^j$$

- Recall: $g^j(x)$ returns a signed real-valued value which can be interpreted as the distance from the separation (hyper)plane to the point x
- Value can also be interpreted as a confidence value. The larger the value the more confident one is that the point x belong to the positive class
- Hence, assign point x to the class whose confidence value is largest for this point

SVM Pairwise classification

- Train a classifier for each possible pair of classes
- For M classes, this results in $\binom{M}{2} = \frac{M \cdot (M-1)}{2}$ binary classifiers
- Classify an unknown point x by applying each of the $\binom{M}{2}$ binary classifiers and count how many times point x was assigned to that class label
- Class label with highest count is then the considered label for the unknown point x

Model selection using evaluation metrics

- Train/test on same data
  - Single metric
  - Typically overfits and likely won't generalize well to new data
  - But can serve as a sanity check: low accuracy on the training set may indicate an implementation problem.
- Single train/test split
  - Single metric
  - Speed and simplicity
  - Lack of variance information
- K-fold cross-validation
  - K train-test splits
  - Average metric over all splits.
  - Can be combined with parameter grid search: GridSearchCV (def. cv = 3)

Training, validation and test framework

- Using only cross-validation or a test set to do model selection may lead to more subtle overfitting / optimistic generalization estimates
- Instead, use three data splits
  - Training set (model building)
  - Validation set (model selection)
  - Test set (final evaluation)
- In practice
  - Create an initial training/test split
  - Do cross-validation on the training data for model/parameter selection
  - Save the held-out test set for final model evaluation

Notes

- Accuracy is often not the right evaluation metric for many real- world machine learning tasks
  - False positives and false negatives may need to be treated very differently
  - Make sure you understand the needs of your application and choose an evaluation metric that matches your application, user, or business goals.
- Examples of additional evaluation methods include
  - Learning curve: How much does accuracy (or other metric) change as a function of the amount of training data?
  - Sensitivity analysis: How much does accuracy (or other metric) change as a function of key learning parameter values?

Machine Learning

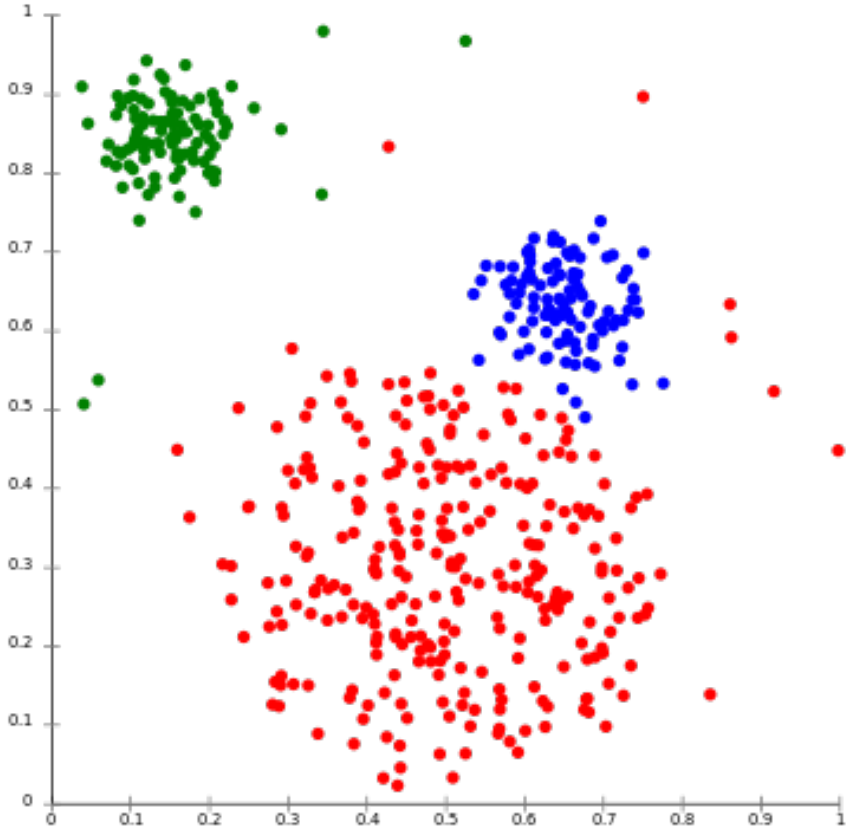
- Supervised learning
- Unsupervised learning: use data ***without*** labels
  - Clustering
    - K-means clustering

Clustering

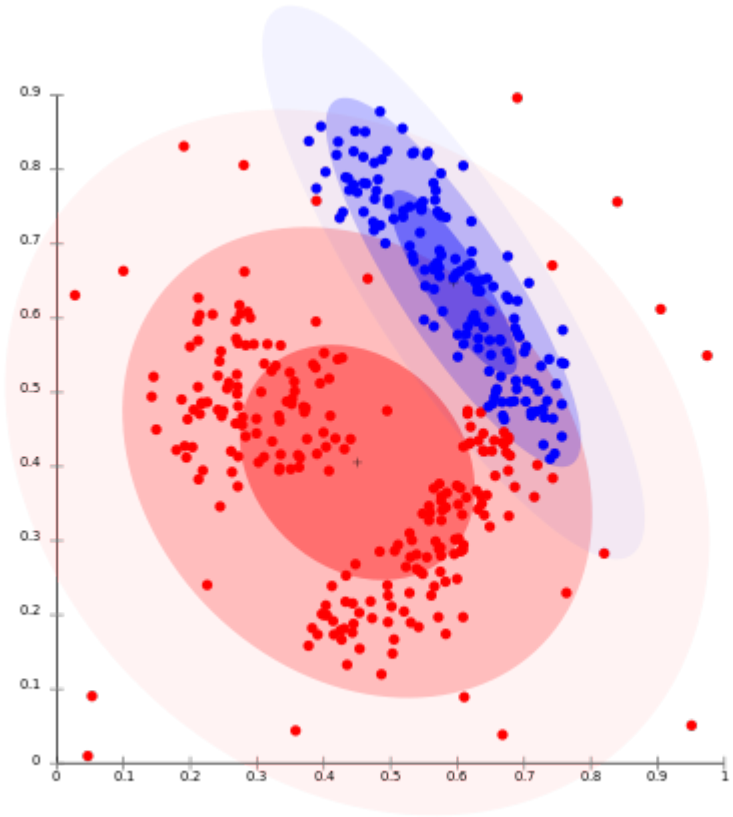
- Clustering
- Applications
  - Market segmentation
  - Social network analysis
  - Astronomical data analysis

Stereotypical Clustering

Points are samples plotted in feature space, e.g., 10,000-dimensional space for 100x100 images.

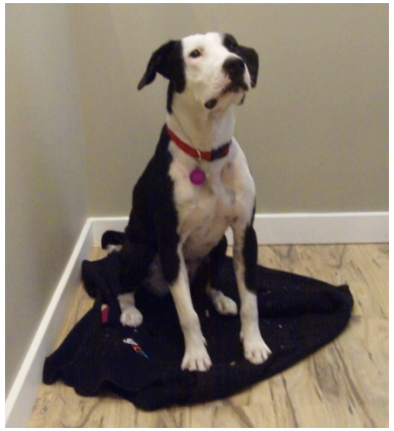


Model-based Clustering



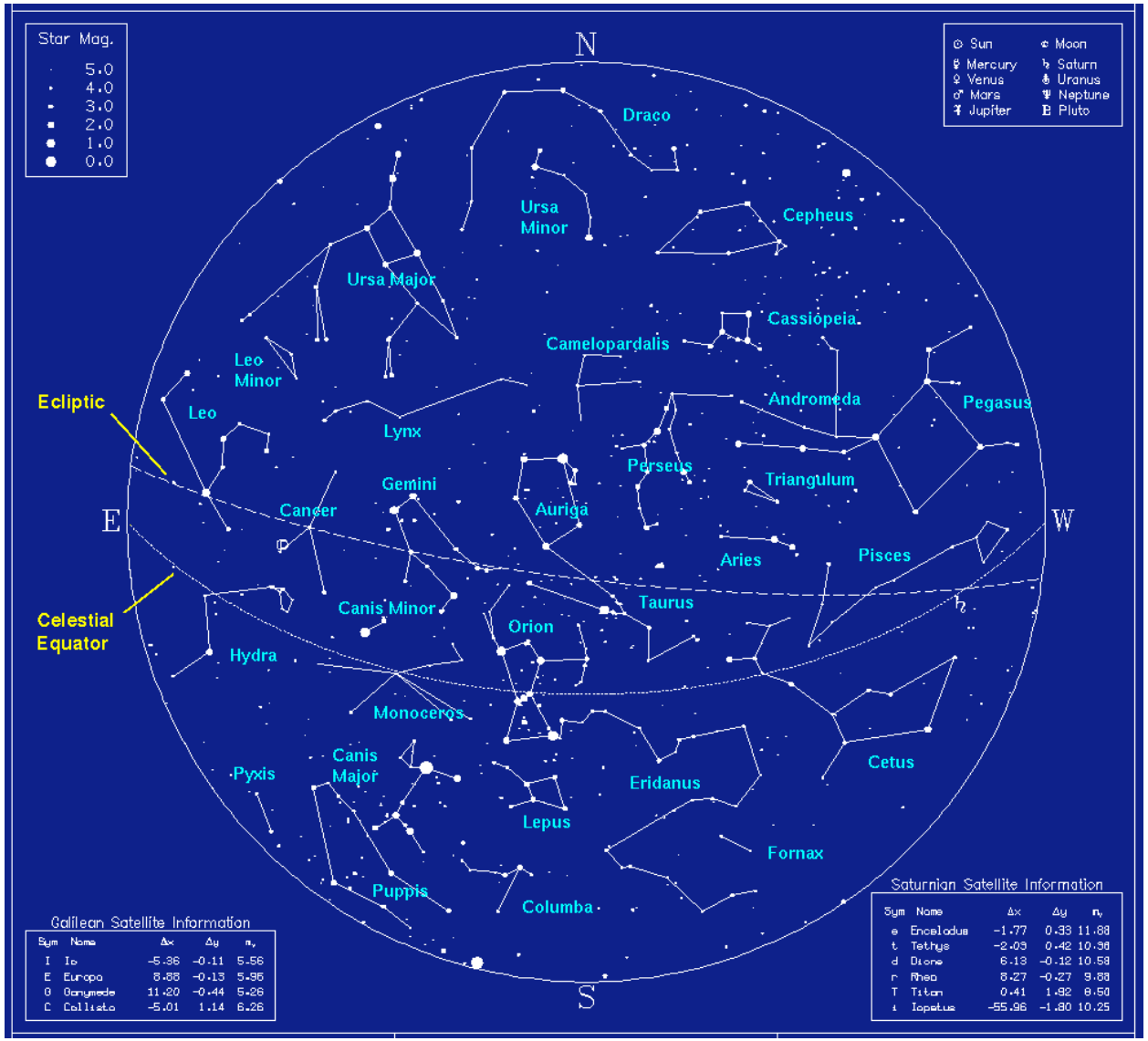
Cluster Bias

- Human beings conceptualize the world through categories represented as *examplars* (Rosch 73, Estes 94).



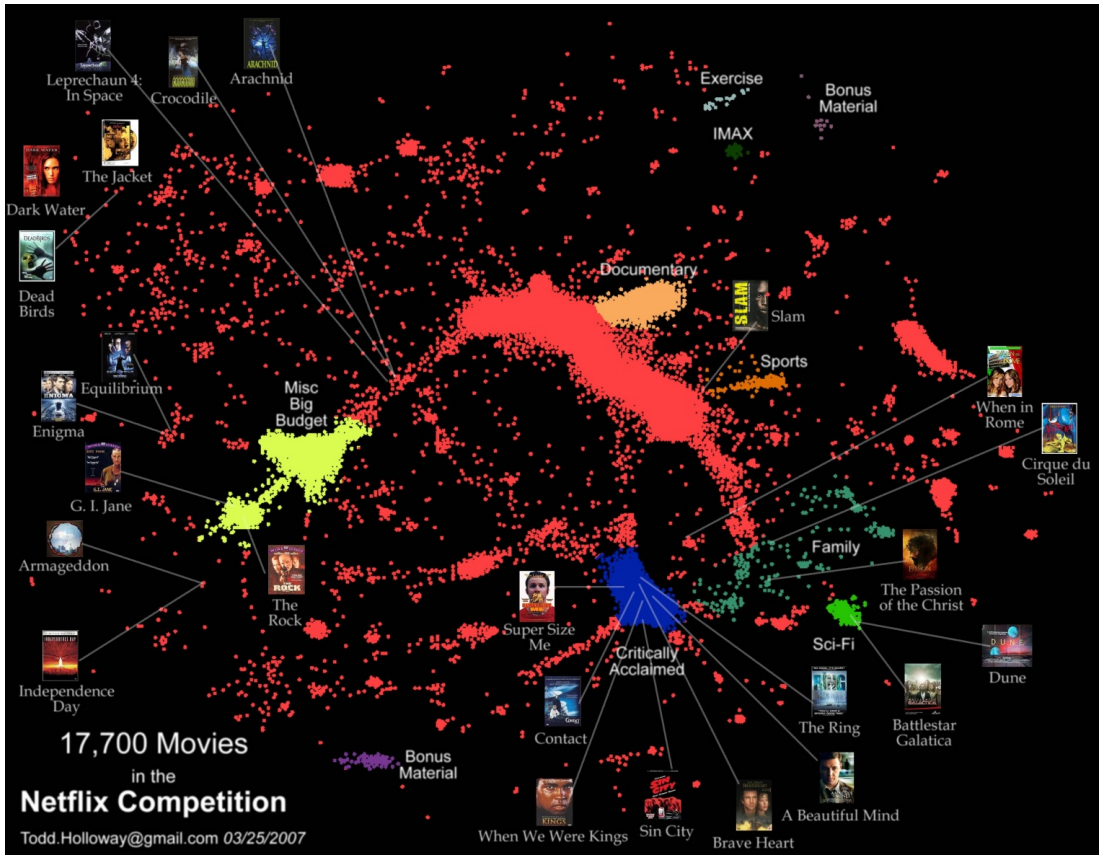
- We tend to see cluster structure whether it is there or not.
- Works well for dogs, but…

Example: Cluster Bias



Example: Netflix

- More of a continuum than discrete clusters
- Factor models, kNN do much better than discrete cluster models.



Cluster Bias

- **Clustering is used more than it should be**, because people assume an underlying domain has discrete classes in it
- This is especially true for characteristics of people, e.g., Myers-Briggs personality types like “ENTP”
- In reality the underlying data is usually continuous
- Just as with Netflix, continuous models (dimension reduction, kNN) tend to do better

Terminology

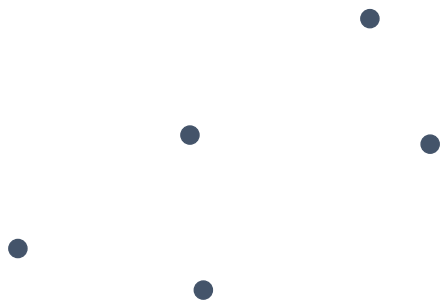
- Hierarchical v.s. flat
  - Hierarchical clustering: clusters form a hierarchy. Can be computed bottom-up or top-down
  - Flat clustering: no inter-cluster structure
- Hard v.s. Soft
  - Hard clustering: items assigned to a unique cluster
  - Soft clustering: cluster membership is a real-valued function, distributed across several clusters

K-means clustering

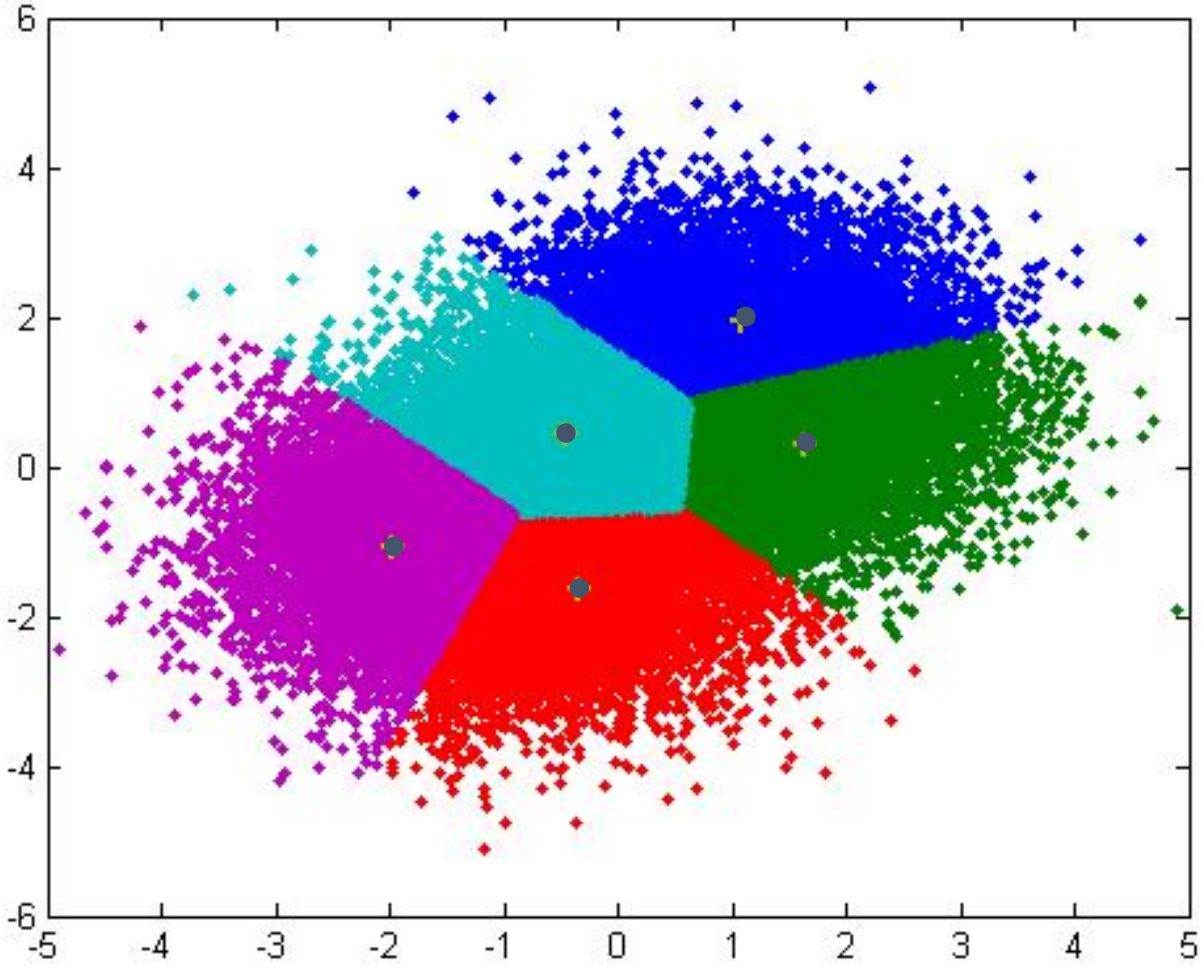
- The standard k-means algorithm is based on Euclidean distance
- The cluster quality measure is an intra-cluster measure only, equivalent to the sum of item-to-centroid kernels
- A simple ***greedy and iterative*** algorithm locally optimizes this measure (usually called Lloyd's algorithm)
  - Find the closest cluster center for each item, and assign it to that cluster
  - Recompute the cluster centroid as the mean of items, for the newly-assigned items in the cluster

K-means clustering

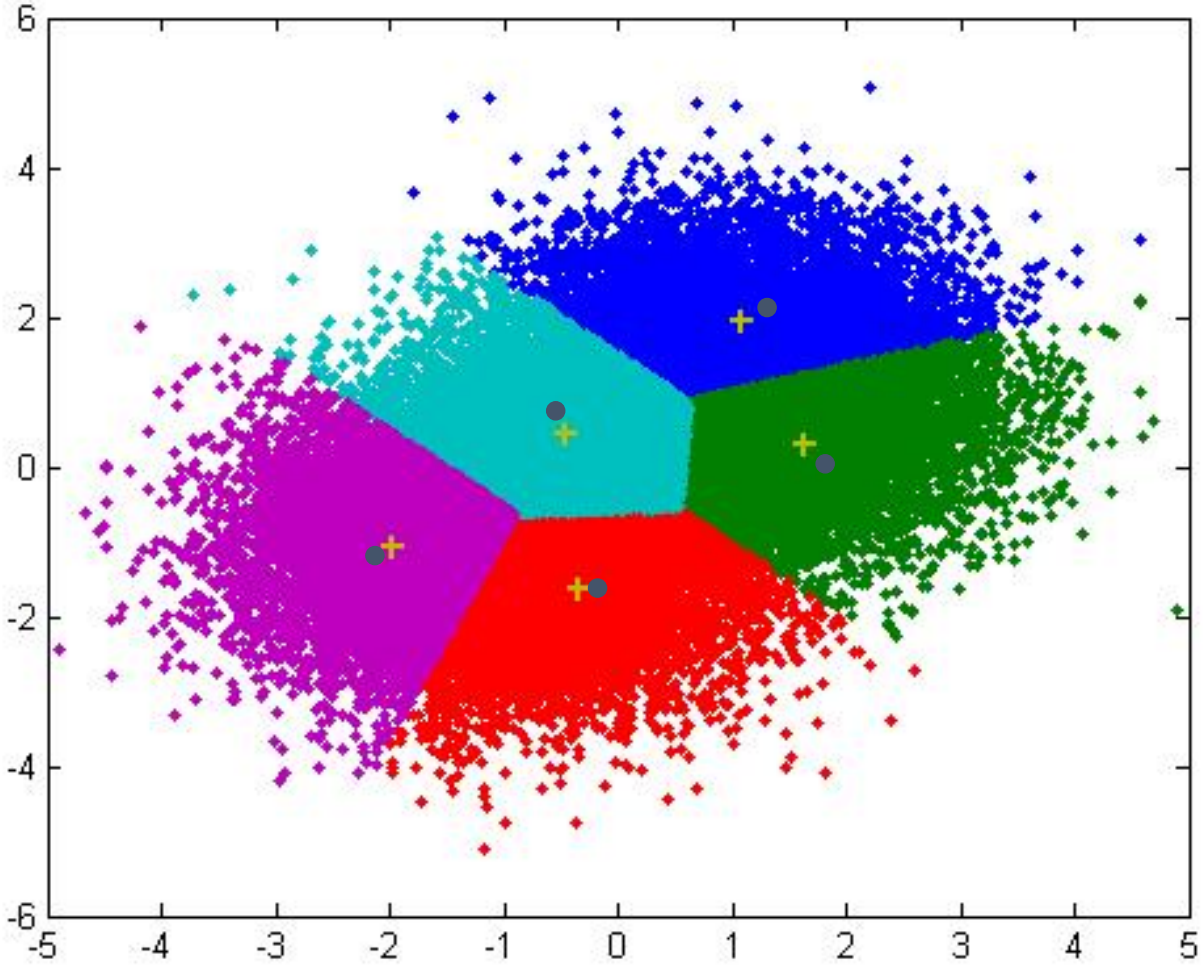
Cluster centers – can pick by sampling the input data.



K-means clustering: assigning points to closest center



K-means clustering: recompute centers*



* cross: old center, circle: new center

K-means clustering

Iterate:

- For fixed number of iterations
- Until no change in assignments
- Until small change in quality

K-means properties

- It's a greedy algorithm with random setup – solution is NOT optimal and varies significantly with different initial points
- Very simple convergence proofs
- Performance is O(nk) per iteration, not bad and can be heuristically improved
  - n = total features in the dataset, k = number clusters
- Many generalizations, e.g.,
  - Fixed-size clusters
  - Simple generalization to m-best soft clustering
- As a "local" clustering method, it works well for data condensation/compression

A few notes on machine learning

- There are numerous algorithms on machine learning
  - You need to understand when they can be applied and what limitations they have
- Do not blindly apply machine learning methods to your data, i.e., you need to keep asking questions on
  - The meaning of the analysis; what it explains
  - Whether you have applied the proper model
  - The further analysis
- Study “deep learning” if you are interested in this area!!!

Additional materials on Machine Learning

- Machine Learning Course for Senior (code-share)
- A tutorial on statistical-learning for scientific data processing (http://scikit-learn.org/stable/tutorial/statistical_inference/index.html)
- Machine Learning on Coursera by Andrew Ng (https://www.coursera.org/learn/machine-learning)
- 모두를 위한 머신러닝/딥러닝 by 김성훈 (https://hunkim.github.io/ml/)
- A large number of books on Machine Learning & Deep Learning

ANY QUESTIONS?